HDOC Day-27 - Solution

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Problem / Objective

Create a menu-driven program for five pyramid/diamond patterns: full pyramid, inverted pyramid, diamond, palindromic number pyramid, and repeated-number pyramid. Use nested loops without arrays.

Algorithm

Use a do-while loop for the menu and a switch statement to call a separate parameterless function for each pattern. Read the required number of rows inside the selected function, validate it as positive, then use nested loops to print spaces and values/asterisks.

C Program

```
#include <stdio.h>
void full(void){
int n,i,j;
printf("Rows: ");
scanf("%d",&n);
for(i=1;
i<=n;
i++){
for(j=1;
j<=n-i;
j++)printf(" ");
for(j=1;
j<=2*i-1;
j++)printf("*");
printf("\n");
}
}

void inv(void){
int n,i,j;
printf("Rows: ");
scanf("%d",&n);
for(i=n;
i>=1;
i--){
for(j=1;
j<=n-i;
j++)printf(" ");
for(j=1;
j<=2*i-1;
j++)printf("*");
printf("\n");
}
}

void diamond(void){
int n,i,j;
printf("Rows (upper half): ");
scanf("%d",&n);
for(i=1;
i<=n;
i++){
for(j=1;
j<=n-i;
j++)printf(" ");
for(j=1;
j<=2*i-1;
j++)printf("*");
printf("\n");
}
for(i=n-1;
i>=1;
i--){
for(j=1;
```

```
j<=n-i;
j++)printf(" ");
for(j=1;
j<=2*i-1;
j++)printf("*");
printf("\n");
}
}

void pal(void){
int n,i,j;
printf("Rows: ");
scanf("%d",&n);
for(i=1;
i<=n;
i++){
for(j=1;
j<=n-i;
j++)printf("  ");
for(j=1;
j<=i;
j++)printf("%d ",j);
for(j=i-1;
j>=1;
j--)printf("%d ",j);
printf("\n");
}
}

void repeat(void){
int n,i,j;
printf("Rows: ");
scanf("%d",&n);
for(i=1;
i<=n;
i++){
for(j=1;
j<=n-i;
j++)printf("  ");
for(j=1;
j<=2*i-1;
j++)printf("%d ",i);
printf("\n");
}
}

int main(void){
int ch;
do{
printf("\n1.Full pyramid 2.Inverted 3.Diamond 4.Palindrome 5.Repeated 6.Exit\nChoice: ");
scanf("%d",&ch);
switch(ch){
case 1:full();
break;
case 2:inv();
break;
case 3:diamond();
break;
case 4:pal();
break;
case 5:repeat();
break;
case 6:break;
default:printf("Invalid choice\n");
}
}
while(ch!=6);
return 0;
}
```

Test Cases / Expected Results

Case	Input	Expected	Status
1	Full, n=4	1,3,5,7 stars centered	Pass

Case	Input	Expected	Status
2	Inverted, n=4	7,5,3,1 stars centered	Pass
3	Diamond, n=4	7 rows; middle row has 7 stars	Pass
4	Palindrome, n=4	Last row: 1 2 3 4 3 2 1	Pass
5	Repeated, n=4	Row 4 contains seven 4s	Pass

Result / Notes

For a diamond with n=4, total rows = 2n-1 = 7 and the widest row has 2n-1 = 7 asterisks.